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# Position-Specific Longitudinal Assessment of Total and Regional Body Composition in NCAA Division I Football Players

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**Abstract**

Brooks, SG and Dengel, DR. Position-specific longitudinal assessment of total and regional body composition in NCAA Division I football players. *J Strength Cond Res* XX(X): 000–000, 2025—Position-specific differences in body composition in American football players have been established. How the total and regional composition changes from preseason to postseason by position is unknown. Thus, this study examined longitudinal changes in total and regional body composition in NCAA Division I collegiate football players (CFPs) over a competitive season. A total of 188 male CFPs aged 18–23 (20.7 ± 1.6 years) were included. Body composition was measured using dual X-ray absorptiometry at preseason and postseason. Subjects were grouped into defensive backs, defensive linemen, linebackers, offensive linemen, punters/kickers, quarterbacks, running backs, tight ends, and wide receivers. Repeated-measures ANOVA assessed effects of season and position on total and regional body composition, controlling for player participation status. Post hoc pairwise comparisons determined the magnitude, direction, and significance of changes for each position. There were no changes in total or regional measures of body mass, fat mass, or bone mass across the season, except for arm mass. Significant ($p < 0.05$) decreases in total lean mass (LM), arm LM, leg LM, and gynoid LM were observed in certain position groups. The results of this study confirm that there is a decrease in total LM during a competitive season in CFPs. This study demonstrates that this loss in LM mainly comes from the leg and gynoid regions of the body and occurs across all positions. This information can be used by strength and nutrition staff to develop new strategies to prevent this LM loss during the season.

**Key Words:** lean mass, fat mass, muscle-to-bone ratio, dual X-ray absorptiometry, athlete

## Introduction

The quantification of body composition provides insight into health and performance in athletics. American football is a unique population to investigate body composition, as the diverse physical demands of each position require different compositions for success. Previously, dual X-ray absorptiometry (DXA) has been utilized to quantify total and regional composition between positions in collegiate and professional football (2–4,18,19,21). Offensive and defensive positions that mirror each other have similar physical demands and thus share similar composition characteristics (3,4). The muscle-to-bone ratio (MBR) has also been used to investigate body composition in football athletes, with significant variations between the various positions but similarities between offensive and defensive positions that mirror each other (5,6).

Skill players (i.e., wide receivers [WRs], running backs [RBs], and defensive backs [DBs]) require speed and agility and, therefore, are typically the smallest in size and lowest in percent body fat (%BF) and strength (3,4,14). Linemen, including offensive linemen (OLs) and defensive linemen (DLs), largely rely on power and mass to tackle and block their opponents (14). For this reason, they are typically the largest in total body mass, highest in % BF, total MBR, and strength, and lowest in speed (2–4,18). Big Skill players (i.e., linebackers [LBs], tight ends [TEs], quarterbacks [QBs], punter/kickers [PKs], and long snappers) act as a transition between the 2 groups, with lean mass (LM) similar to skill players, fat mass (FM) similar to linemen, and intermediate performance in strength and speed (3,12,14,19).

Previous longitudinal studies have demonstrated seasonal fluctuations in body composition among collegiate football players (CFPs), including reductions in LM and increases in FM during the competitive season (1,19). These changes may be attributed to shifts in training emphasis across seasonal phases, particularly during in-season periods when training intensity often declines to prioritize game performance and recovery (9,16). However, to date, longitudinal analyses of body composition in football athletes are limited in sample size, broad grouping of positions, and focuse on total body composition without regional differentiation (1,12,16,17,19). Therefore, there is a need for understanding how regional and total body composition changes over time by position.

The purpose of this investigation was to determine how total and regional body composition changes during a competitive collegiate football season in individual positions. It was hypothesized that significant changes in total and regional body composition would occur from preseason to postseason. Specifically, it was hypothesized that LM would decrease during the competitive season. In addition, it was hypothesized that body composition and the magnitude of seasonal changes would vary between positions. The secondary purpose of this investigation was to determine how total and regional MBR change during the competitive season. It was hypothesized that significant changes

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in ratios would occur from preseason to postseason, with positional variations in magnitude.

## Methods

### Experimental Approach to the Problem

This longitudinal study utilized previously collected data from a larger sample of Division I CFPs. The sample consisted of 1,091 scans from 275 subjects collected between 2019 and 2023. Dual X-ray absorptiometry was used to measure total and regional body composition at preseason and postseason timepoints. Differences in total and regional body composition variables were analyzed from preseason to postseason. Only 1 set of scans for each player was used in the analysis, resulting in a sample size of 188 subjects and 376 scans.

### Subjects

A total of 188 male CFPs aged 18–23 (20.7 ± 1.6 years) were included in this study. Subjects were grouped positionally into DBs, DLs, LBs, OLs, PKs, QBs, RBs, TEs, and WRs. The inclusion criteria were subjects with both preseason and postseason visits from the same competitive season. For subjects with both visits in more than 1 season, the most recent season was used. All preseason scans were completed during a 6-week period in June and July. Postseason scans began immediately following the last competitive game in January and were completed within a 2-week period. These time periods were consistent across all subjects and years of data collection. Owing to scheduling constraints, subjects may not have been scanned at the same time of day for the preseason and postseason scans; however, the same protocols were followed, including instructions for subjects to arrive fasted and well hydrated. The exclusion criteria were an incomplete pair of scans falling in the defined periods from the same competitive season. Before participation, written consent was obtained from each athlete. This study was conducted in accordance with the Declaration of Helsinki of 1975, and approval for retrospective statistical analysis of preexisting DXA scan data was given by The University of Minnesota's Institutional Review Board.

### Procedures

A whole body DXA (Hologic Horizon A; Hologic Inc., Marlborough, MA) scan was performed, measuring FM, LM, and bone mineral content (BMC). Lean mass includes muscle mass, organs, total body water, and fat inseparable from these structures. Standard DXA protocols were followed, with subjects instructed to arrive well hydrated and fasted. Before scanning, subjects were instructed to remove all metal objects, thick clothing, and heavy plastic to reduce interference with the scan. On the day of the scan, age, height, body mass, and playing position were recorded. Height and body mass were collected using a stadiometer and a digital scale, respectively. Each subject then lay in a supine position at the center of the scanning table. If subjects were too tall for the table, the head was excluded from the scanning region, as it is estimated and not relevant to the scope of the study. If subjects were too wide for the table, they were positioned so that only an arm would be left out of the scanning region. Hologic reflection software duplicated the arm fully within the scanning area and used the information to replace the arm that was not fully in the scan area. Subjects were scanned at The University of Minnesota Twin Cities, and all scan files were analyzed using Hologic Apex software (General Electric Medical System Apex version 5.6.0.4, Hologic Inc.) to determine relative fat (%BF), total LM, total FM, BMC, bone mineral density (BMD), and visceral adipose tissue (VAT) mass. Regional measures were also determined, including LM, FM, BMC, and BMD for arm, leg, and trunk regions, with trunk measured from the chin to the top of the iliac crest. In addition, FM and LM measures were determined for the android (area between ribs and pelvis) and gynoid (region including hips and upper thighs) regions (7).

### Statistical Analyses

All analyses were performed using RStudio (version 4.4.2). Muscle-to-bone ratio was calculated totally and regionally during preseason and postseason using LM/BMC. Two-factor repeated-measures ANOVA assessed effects of season and playing position on total and regional body composition and MBR, controlling for playing status as a covariate. Playing status was determined using a grouping method with the following categories: injured for the season, did not play (did not play in at least 25% of games, including redshirts), played (played in at least 25% of games), and started (played a majority of games and started at least half). Bonferroni-corrected post hoc comparisons determined the magnitude, direction, and significance ($p < 0.05$) of changes within each position group. All visualizations were created using GraphPad PRISM (version 10.1.2).

## Results

Comparisons of physical characteristics [i.e., age, height, mass] of study subjects at preseason are presented in Table 1 by position. Offensive linemen were significantly taller than all positions except TEs. However, TEs were taller than all other positions except DLs. In addition, DLs were taller than LBs, PKs, WRs, DBs, and RBs, but not QBs. By contrast, DBs and RBs were shorter than all other positions. Offensive linemen were significantly heavier than all other positions. Defensive linemen were heavier than TEs. In

**Table 1**
**Physical characteristics (mean ± *SD*) of study subject.*†**

| | Total | DB | DL | LB | OL | PK | QB | RB | TE | WR |
|---|---|---|---|---|---|---|---|---|---|---|
| n | 188 | 34 | 28 | 25 | 27 | 10 | 10 | 13 | 14 | 27 |
| Age (y) | 20.7 ± 1.4 | 20.7 ± 1.1[a] | 20.6 ± 1.4[a] | 20.6 ± 1.3[a] | 20.9 ± 1.6[a] | 21.0 ± 1.3[a] | 20.3 ± 1.6[a] | 20.2 ± 1.9[a] | 20.9 ± 1.3[a] | 20.6 ± 1.5[a] |
| Height (cm) | 187.5 ± 22.8 | 181.1 ± 5.7[a] | 191.4 ± 3.8[bc] | 187.3 ± 3.2[d] | 197.0 ± 3.1[e] | 187.4 ± 7.2[d] | 188.4 ± 3.8[bd] | 179.0 ± 4.6[a] | 194.2 ± 2.5[ce] | 185.1 ± 5.8[d] |
| Mass (kg) | 187.9 ± 7.2 | 88.5 ± 9.1[a] | 127.7 ± 13.8[b] | 103.9 ± 5.7[c] | 145.4 ± 13.9[d] | 95.0 ± 8.4[a] | 94.3 ± 8.0[a] | 92.1 ± 5.9[a] | 115.6 ± 6.5[e] | 88.6 ± 7.7[a] |

*DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver.
†Different letters denote significant ($p < 0.05$) differences between position groups.



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**Table 2**

**Seasonal changes (mean ± *SD*) in total body composition variables by position.*†**

| | DB (*n* = 34) | DL (*n* = 28) | LB (*n* = 25) | OL (*n* = 27) | PK (*n* = 10) | QB (*n* = 10) | RB (*n* = 13) | TE (*n* = 14) | WR (*n* = 27) |
|---|---|---|---|---|---|---|---|---|---|
| Total mass (kg) | | | | | | | | | |
| Pre | 89.85 ± 8.91 | 128.08 ± 14.36 | 104.82 ± 5.95 | 145.49 ± 15.96 | 95.87 ± 8.53 | 95.34 ± 8.83 | 93.55 ± 6.82 | 116.22 ± 6.40 | 89.89 ± 7.94 |
| Post | 89.30 ± 8.76 | 128.06 ± 13.85 | 104.94 ± 5.70 | 143.04 ± 13.07 | 95.16 ± 8.12 | 95.50 ± 7.40 | 93.39 ± 5.80 | 115.23 ± 5.64 | 89.19 ± 7.71 |
| Total LM (kg) | | | | | | | | | |
| Pre | 76.17 ± 7.03 | 97.53 ± 6.09 | 86.09 ± 5.38 | 104.00 ± 6.96 | 78.53 ± 6.57 | 79.24 ± 7.21 | 78.60 ± 4.99 | 92.80 ± 5.23 | 76.90 ± 5.95 |
| Post | 75.12 ± 6.50** | 96.80 ± 6.23 | 85.60 ± 5.02 | 102.30 ± 6.15*** | 76.92 ± 6.45* | 78.01 ± 5.76 | 77.30 ± 3.83* | 91.39 ± 4.36* | 75.73 ± 5.67* |
| Total FM (kg) | | | | | | | | | |
| Pre | 13.67 ± 2.51 | 30.55 ± 9.15 | 18.73 ± 2.83 | 41.49 ± 11.09 | 17.34 ± 3.47 | 16.09 ± 3.02 | 14.95 ± 3.11 | 23.42 ± 4.50 | 12.99 ± 2.54 |
| Post | 14.18 ± 3.00 | 31.26 ± 9.04 | 19.34 ± 3.01 | 40.69 ± 9.72 | 18.25 ± 3.78 | 17.49 ± 2.87 | 16.09 ± 3.61 | 23.83 ± 4.58 | 13.46 ± 2.72 |
| Total BMC (kg) | | | | | | | | | |
| Pre | 10.57 ± 1.12 | 13.36 ± 1.13 | 11.87 ± 1.20 | 13.94 ± 1.64 | 10.29 ± 1.07 | 10.39 ± 1.11 | 10.84 ± 0.99 | 12.28 ± 1.10 | 10.38 ± 1.03 |
| Post | 10.28 ± 1.02 | 13.34 ± 1.10 | 11.75 ± 1.00 | 13.70 ± 1.43 | 10.05 ± 1.01 | 10.05 ± 0.92 | 10.63 ± 1.02 | 12.00 ± 1.09 | 10.27 ± 0.99 |

*LM = lean mass; FM = fat mass; BMC = bone mineral content; DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver.
†Significance levels: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.

addition, TEs were heavier than LBs. However, LBs were heavier than DBs, PKs, QBs, RBs, and WRs.

Position was a significant factor for all body composition variables, meaning that position significantly influenced body composition. Total body composition at preseason and postseason is summarized by position in Table 2. Totally, there were no significant seasonal changes in total body mass, FM, BMC, or BMD; however, LM showed a significant ($p < 0.01$) change with a medium effect size. Total LM (kg) decreased ($p < 0.05$) from preseason to postseason in DBs, OLs, PKs, RBs, TEs, and WRs. However, there were no significant changes in total LM in DLs, LBs, or QBs (Figure 1A).

Preseason and postseason regional body composition (arms, trunk, legs) are presented in Tables 3–5. Regionally, there were no significant seasonal changes in regional body mass, FM, or BMC, except for arm mass. Arm mass significantly ($p < 0.05$) decreased in OLs, but did not change in any other position group. In DBs, arm LM (kg) decreased between seasons with a small effect size ($p < 0.05$). However, there was no change in arm LM in DLs, LBs, OLs, PKs, QBs, RBs, TEs, or WRs (Figure 1B). Trunk LM did not significantly change between seasons in any position (Figure 1C). However, leg LM decreased with a large effect size from preseason to postseason in DBs, DLs, OLs, PKs, RBs, TEs, and WRs ($p < 0.05$). There were no significant changes in leg LM

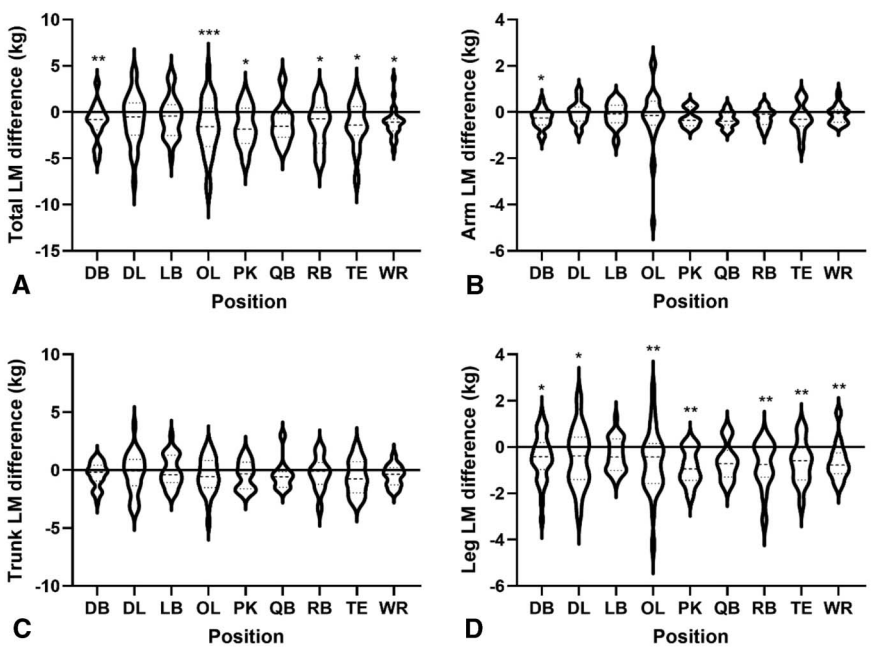


**Figure 1.** Total and regional lean mass changes from preseason to postseason by position. (A) Total lean mass; (B) arm lean mass; (C) trunk lean mass; (D) leg lean mass. DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver. Significance levels: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.



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**Table 3**
**Seasonal changes (mean ± *SD*) in arm composition by position.*†**

| | DB (*n* = 34) | DL (*n* = 28) | LB (*n* = 25) | OL (*n* = 27) | PK (*n* = 10) | QB (*n* = 10) | RB (*n* = 13) | TE (*n* = 14) | WR (*n* = 27) |
|---|---|---|---|---|---|---|---|---|---|
| Arm mass (kg) | | | | | | | | | |
| Pre | 12.32 ± 1.34 | 17.01 ± 2.00 | 14.23 ± 1.30 | 18.86 ± 2.43 | 12.40 ± 1.29 | 12.42 ± 1.36 | 12.71 ± 1.22 | 15.07 ± 1.03 | 11.99 ± 1.24 |
| Post | 12.07 ± 1.24 | 17.09 ± 1.90 | 14.22 ± 1.03 | 18.52 ± 1.67* | 12.34 ± 1.21 | 12.30 ± 1.31 | 12.67 ± 1.11 | 14.88 ± 1.02 | 11.99 ± 1.22 |
| Arm LM (kg) | | | | | | | | | |
| Pre | 10.57 ± 1.12 | 13.36 ± 1.13 | 11.87 ± 1.20 | 13.94 ± 1.64 | 10.29 ± 1.07 | 10.39 ± 1.11 | 10.84 ± 0.99 | 12.28 ± 1.10 | 10.38 ± 1.03 |
| Post | 10.28 ± 1.02* | 13.34 ± 1.10 | 11.75 ± 1.00 | 13.70 ± 1.43 | 10.05 ± 1.01 | 10.05 ± 0.92 | 10.63 ± 1.02 | 12.00 ± 1.09 | 10.27 ± 0.99 |
| Arm FM (kg) | | | | | | | | | |
| Pre | 1.75 ± 0.29 | 3.65 ± 1.15 | 2.37 ± 0.38 | 4.91 ± 1.13 | 2.11 ± 0.44 | 2.03 ± 0.46 | 1.86 ± 0.40 | 2.79 ± 0.50 | 1.62 ± 0.27 |
| Post | 1.79 ± 0.31 | 3.74 ± 1.10 | 2.46 ± 0.37 | 4.82 ± 0.76 | 2.29 ± 0.45 | 2.25 ± 0.50 | 2.04 ± 0.36 | 2.88 ± 0.58 | 1.72 ± 0.30 |
| Arm BMC (kg) | | | | | | | | | |
| Pre | 0.57 ± 0.06 | 0.67 ± 0.07 | 0.61 ± 0.06 | 0.68 ± 0.07 | 0.51 ± 0.08 | 0.55 ± 0.06 | 0.55 ± 0.05 | 0.65 ± 0.07 | 0.56 ± 0.07 |
| Post | 0.57 ± 0.06 | 0.69 ± 0.07 | 0.62 ± 0.06 | 0.68 ± 0.08 | 0.52 ± 0.08 | 0.56 ± 0.06 | 0.56 ± 0.05 | 0.66 ± 0.06 | 0.57 ± 0.07 |

*LM = lean mass; FM = fat mass; BMC = bone mineral content; BMD = bone mineral density; DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver.
†Significance levels: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.

in LBs or QBs (Figure 1D). Regionally, BMD did not change between seasons in the arm, trunk, or leg. Pelvis BMD significantly ($p < 0.05$) changed from preseason to postseason with a medium effect size, increasing in DBs, DLs, OLs, TEs, and WRs.

Android and gynoid composition at preseason and postseason are presented by position in Table 6. There were no significant changes in android FM (Figure 2A) or android LM (Figure 2B). Gynoid FM did not change between seasons (Figure 2C). Gynoid LM significantly ($p < 0.001$) decreased between seasons with a large effect in all positions but QBs (Figure 2D). Visceral adipose tissue mass did not significantly change between seasons (Table 6).

Total and regional MBR at preseason and postseason are presented by position in Table 7. Total MBR significantly ($p < 0.01$) changed from preseason to postseason with a large effect size, decreasing in DBs, DLs, TEs, and WRs. Total MBR did not change in LBs, OLs, PKs, QBs, or RBs (Figure 3A). Arm MBR significantly ($p < 0.05$) changed between seasons with a large effect size, decreasing in DBs, DLs, LBs, PKs, QBs, RBs, TEs, and WRs. Arm MBR did not change in OLs (Figure 3B). Trunk MBR decreased from preseason to postseason with a small effect size in DBs ($p < 0.05$). There were no significant changes in trunk MBR in DLs, LBs, OLs, PKs, QBs, RBs, TEs, or WRs (Figure 3C). Leg MBR significantly ($p < 0.01$) changed with a large effect size, decreasing in DBs, DLs, LBs, OLs, PKs, QBs, RBs, TEs, and WRs (Figure 3D).

## Discussion

Longitudinal changes in body composition in CFPs have previously been studied (1,8,12,16,17,19). However, this is the first study to examine position-specific longitudinal changes to regional body composition in CFPs. In addition, this study is the first to longitudinally investigate both total and regional MBR in CFPs. Notable observations include a loss of LM across the season, while body mass, FM, and bone mass remained relatively stable.

There were no significant seasonal changes in total body mass, FM, or BMC; however, total LM significantly decreased in every position except LBs and QBs. These findings align with previous literature showing seasonal LM losses in CFPs (1,19), although some studies have reported LM gains over the season (17). This discrepancy may be due to a difference in the timing of seasonal assessments. For example, the study by Trexler et al. (17) measured preseason composition sooner and postseason competition later than other studies (1,19), potentially capturing LM gains from the off-season. The absence of total LM losses in LBs and QBs, both positions classified as Big Skill, differs from previous work that observed significant declines in LM in the Big Skill group (19). This may be due to the broad grouping of positions and the lack of repeated-measure analysis in previous research (1,19). This study's more specific positional analysis suggests that Big Skill athletes may not uniformly lose LM, highlighting the importance of individualized assessment.

**Table 4**
**Seasonal changes (mean ± *SD*) in trunk composition by position.***

| | DB (*n* = 34) | DL (*n* = 28) | LB (*n* = 25) | OL (*n* = 27) | PK (*n* = 10) | QB (*n* = 10) | RB (*n* = 13) | TE (*n* = 14) | WR (*n* = 27) |
|---|---|---|---|---|---|---|---|---|---|
| Trunk mass (kg) | | | | | | | | | |
| Pre | 39.69 ± 4.28 | 58.99 ± 8.40 | 47.26 ± 3.00 | 70.20 ± 8.81 | 44.45 ± 5.13 | 44.90 ± 4.18 | 42.01 ± 3.72 | 54.26 ± 3.73 | 39.82 ± 3.84 |
| Post | 39.58 ± 4.15 | 59.02 ± 7.91 | 47.56 ± 2.91 | 69.36 ± 8.04 | 44.52 ± 5.47 | 45.03 ± 3.42 | 42.15 ± 3.10 | 53.79 ± 3.33 | 39.59 ± 3.97 |
| Trunk LM (kg) | | | | | | | | | |
| Pre | 34.25 ± 3.49 | 45.28 ± 3.56 | 39.49 ± 2.40 | 50.09 ± 3.63 | 37.04 ± 3.62 | 37.91 ± 3.50 | 36.01 ± 2.81 | 44.33 ± 2.92 | 34.76 ± 3.02 |
| Post | 33.88 ± 3.17 | 45.00 ± 3.52 | 39.44 ± 2.25 | 49.65 ± 3.44 | 36.60 ± 4.03 | 37.50 ± 2.79 | 35.73 ± 2.05 | 43.72 ± 2.56 | 34.29 ± 3.07 |
| Trunk FM (kg) | | | | | | | | | |
| Pre | 5.44 ± 1.10 | 13.71 ± 5.30 | 7.76 ± 1.51 | 20.10 ± 6.89 | 7.41 ± 1.87 | 6.99 ± 1.38 | 6.00 ± 1.37 | 9.92 ± 2.11 | 5.06 ± 1.06 |
| Post | 5.71 ± 1.34 | 14.01 ± 5.18 | 8.12 ± 1.58 | 19.71 ± 6.23 | 7.91 ± 2.16 | 7.54 ± 1.11 | 6.41 ± 1.66 | 10.07 ± 2.01 | 5.30 ± 1.16 |
| Trunk BMC (kg) | | | | | | | | | |
| Pre | 1.02 ± 0.11 | 1.17 ± 0.12 | 1.13 ± 0.09 | 1.21 ± 0.13 | 1.01 ± 0.19 | 1.10 ± 0.16 | 1.03 ± 0.11 | 1.22 ± 0.12 | 1.08 ± 0.11 |
| Post | 1.03 ± 0.10 | 1.17 ± 0.12 | 1.12 ± 0.09 | 1.20 ± 0.12 | 1.00 ± 0.19 | 1.11 ± 0.15 | 1.03 ± 0.11 | 1.23 ± 0.13 | 1.08 ± 0.10 |

*LM = lean mass; FM = fat mass; BMC = bone mineral content; BMD = bone mineral density; DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver.



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**Table 5**
**Seasonal changes (mean ± *SD*) in leg composition by position.*†**

| | DB (*n* = 34) | DL (*n* = 28) | LB (*n* = 25) | OL (*n* = 27) | PK (*n* = 10) | QB (*n* = 10) | RB (*n* = 13) | TE (*n* = 14) | WR (*n* = 27) |
|---|---|---|---|---|---|---|---|---|---|
| Leg mass (kg) | | | | | | | | | |
| Pre | 32.77 ± 3.98 | 47.23 ± 5.28 | 38.38 ± 3.04 | 52.54 ± 7.10 | 34.44 ± 2.93 | 33.14 ± 3.66 | 33.40 ± 2.39 | 42.92 ± 3.12 | 33.00 ± 3.44 |
| Post | 32.52 ± 4.17 | 47.09 ± 5.27 | 38.21 ± 2.79 | 51.67 ± 6.31 | 33.84 ± 2.49 | 33.12 ± 3.12 | 33.17 ± 2.34 | 42.40 ± 3.13 | 32.51 ± 3.12 |
| Leg LM (kg) | | | | | | | | | |
| Pre | 27.49 ± 29.84 | 35.21 ± 2.78 | 30.96 ± 2.63 | 37.03 ± 3.65 | 27.72 ± 2.73 | 27.23 ± 3.10 | 27.61 ± 1.79 | 33.17 ± 2.10 | 27.90 ± 2.50 |
| Post | 27.05 ± 2.90* | 34.76 ± 3.02* | 30.64 ± 2.40 | 36.38 ± 3.39** | 26.87 ± 2.51** | 26.61 ± 2.62 | 26.81 ± 1.50** | 32.52 ± 2.17** | 27.29 ± 2.16** |
| Leg FM (kg) | | | | | | | | | |
| Pre | 5.28 ± 1.32 | 12.02 ± 3.31 | 7.42 ± 1.36 | 15.51 ± 4.15 | 6.72 ± 1.42 | 5.91 ± 1.28 | 5.79 ± 1.41 | 9.75 ± 2.20 | 5.10 ± 1.32 |
| Post | 5.47 ± 1.60 | 13.33 ± 3.34 | 7.58 ± 1.32 | 15.29 ± 3.80 | 6.97 ± 1.55 | 6.51 ± 1.37 | 6.35 ± 1.70 | 9.88 ± 2.28 | 5.22 ± 1.40 |
| Leg BMC (kg) | | | | | | | | | |
| Pre | 1.36 ± 0.13 | 1.60 ± 0.15 | 1.47 ± 0.13 | 1.63 ± 0.12 | 1.35 ± 0.19 | 1.36 ± 0.14 | 1.32 ± 0.11 | 1.56 ± 0.14 | 1.40 ± 0.13 |
| Post | 1.38 ± 0.13 | 1.61 ± 0.14 | 1.49 ± 0.13 | 1.63 ± 0.14 | 1.36 ± 0.18 | 1.38 ± 0.15 | 1.33 ± 0.11 | 1.60 ± 0.14 | 1.41 ± 0.13 |

*LM = lean mass; FM = fat mass; BMC = bone mineral content; BMD = bone mineral density; DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver.
†Significance levels: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.

Although this study did not examine player seniority (academic class or years of collegiate eligibility), previous research has noted interactions between seniority and position, with underclassmen linemen significantly gaining FM over the season (1). Although this study did not observe significant FM changes in linemen, this points to the need for more nuanced monitoring strategies during the season and tailored training and nutrition support for the individual player.

There were no significant changes in total FM or in any regional measures (arm, trunk, leg). This contrasts with findings from Binkley et al. (1), who classified positions as linemen and non-linemen and reported FM gains in the legs for both groups, and gains in the arms for linemen. Although Binkley et al. also found LM losses across all regions in both groups, our results provide more position-specific observations. Notably, arm LM declined significantly only in DBs, and trunk LM remained unchanged in all positions. However, leg LM decreased significantly in every position except LBs and QBs. These differences may stem from variations in training and playing demands. The widespread loss of leg LM across most positions suggests there may be a potential decline in lower-body strength and power during the season. As such, greater emphasis on maintaining lower-body LM in-season may be warranted, particularly for positions reliant on explosive movement.

There is yet to be an investigation of how regional BMC and BMD change across a season in CFPs. Wichmann et al. (19) tracked total BMD across a season and found significant decreases from preseason to postseason in the total cohort. By contrast, in this study, total and regional BMC and BMD did not change from preseason to postseason; however, increases in pelvis BMD were observed in many groups. This increase in pelvis BMD may indicate an adaptation to high-impact forces experienced in competition. Notably, bone adaptations have been documented over similar timeframes in athletic populations, with region-specific changes occurring in response to mechanical loading and sport participation (11,15). The difference in observations may be due to the cohort grouping and lack of repeated-measure analysis by Wichmann et al. (19).

**Table 6**
**Seasonal changes (mean ± *SD*) in android and gynoid body composition variables by position.*†**

| | DB (*n* = 34) | DL (*n* = 28) | LB (*n* = 25) | OL (*n* = 27) | PK (*n* = 10) | QB (*n* = 10) | RB (*n* = 13) | TE (*n* = 14) | WR (*n* = 27) |
|---|---|---|---|---|---|---|---|---|---|
| VAT (kg) | | | | | | | | | |
| Pre | 0.32 ± 0.06 | 0.42 ± 0.11 | 0.35 ± 0.07 | 0.54 ± 0.20 | 0.31 ± 0.04 | 0.35 ± 0.08 | 0.34 ± 0.06 | 0.35 ± 0.05 | 0.30 ± 0.04 |
| Post | 0.32 ± 0.04 | 0.43 ± 0.11 | 0.34 ± 0.06 | 0.53 ± 0.15 | 0.31 ± 0.05 | 0.32 ± 0.05 | 0.34 ± 0.06 | 0.36 ± 0.07 | 0.32 ± 0.04 |
| Android LM (kg) | | | | | | | | | |
| Pre | 4.50 ± 0.64 | 6.45 ± 0.87 | 5.25 ± 0.44 | 7.35 ± 0.77 | 5.14 ± 0.74 | 4.98 ± 0.52 | 4.77 ± 0.54 | 6.02 ± 0.42 | 4.48 ± 0.42 |
| Post | 4.42 ± 0.61 | 6.45 ± 0.81 | 5.34 ± 0.38 | 7.28 ± 0.70 | 5.13 ± 0.76 | 5.00 ± 0.48 | 4.72 ± 0.51 | 5.93 ± 0.54 | 4.47 ± 0.45 |
| Android FM (kg) | | | | | | | | | |
| Pre | 0.90 ± 0.20 | 2.45 ± 0.96 | 1.29 ± 0.32 | 3.70 ± 1.32 | 1.24 ± 0.39 | 1.12 ± 0.22 | 0.99 ± 0.21 | 1.68 ± 0.47 | 0.80 ± 0.17 |
| Post | 0.96 ± 0.25 | 2.56 ± 1.03 | 1.38 ± 0.33 | 3.68 ± 1.19 | 1.38 ± 0.50 | 1.18 ± 0.21 | 1.08 ± 0.27 | 1.75 ± 0.50 | 0.84 ± 0.19 |
| Gynoid LM (kg) | | | | | | | | | |
| Pre | 12.53 ± 1.37 | 15.94 ± 1.54 | 14.15 ± 0.89 | 17.56 ± 1.93 | 13.06 ± 1.15 | 12.97 ± 1.47 | 12.81 ± 0.96 | 15.50 ± 1.30 | 12.60 ± 1.06 |
| Post | 12.22 ± 1.26*** | 15.67 ± 1.63** | 13.91 ± 0.82* | 17.22 ± 1.74*** | 12.60 ± 1.01** | 12.69 ± 1.08 | 12.49 ± 0.80* | 15.07 ± 1.22*** | 12.26 ± 1.07*** |
| Gynoid FM (kg) | | | | | | | | | |
| Pre | 2.48 ± 0.56 | 5.51 ± 1.43 | 3.56 ± 0.61 | 7.36 ± 1.94 | 3.32 ± 0.72 | 2.93 ± 0.60 | 0.99 ± 0.21 | 1.68 ± 0.47 | 2.40 ± 0.62 |
| Post | 2.54 ± 0.69 | 5.63 ± 1.42 | 3.71 ± 0.66 | 7.31 ± 1.74 | 3.46 ± 0.80 | 3.21 ± 0.57 | 1.08 ± 0.27 | 1.75 ± 0.50 | 2.45 ± 0.65 |

*VAT = visceral adipose tissue; LM = lean mass; FM = fat mass; DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver.
†Significance levels: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.



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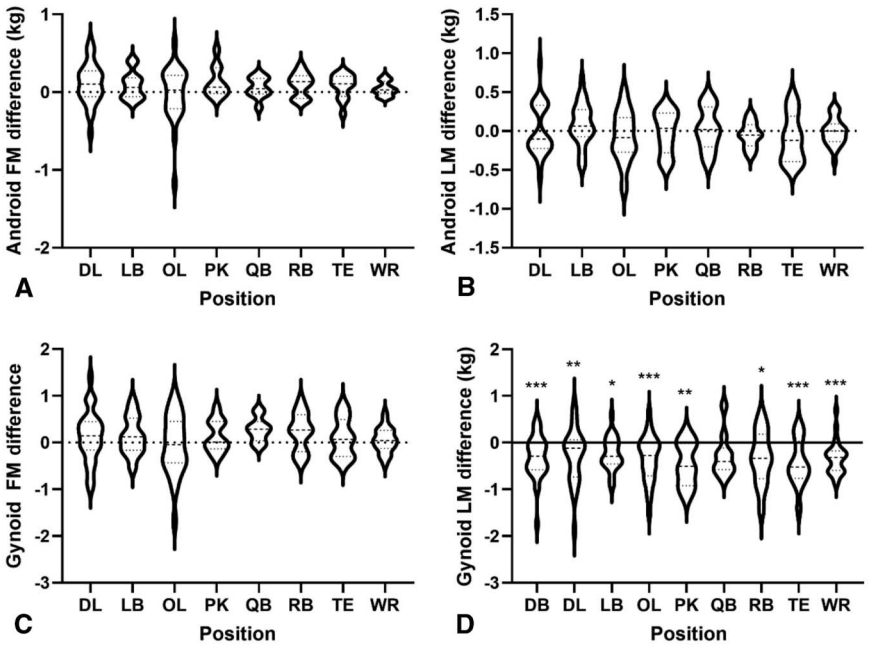


**Figure 2.** Android and gynoid composition changes from preseason to postseason by position. (A) Android fat mass; (B) android lean mass; (C) gynoid fat mass; (D) gynoid lean mass. DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver. Significance levels: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.

This is the first study to present seasonal changes in DXA-derived android and gynoid regions. Android LM and FM, VAT mass, and gynoid FM did not change between seasons in any position. Gynoid LM significantly decreased across the season in all positions except QBs. VAT mass has been previously investigated; however, significant changes have only been observed from postseason to spring season, with no significant changes across the competitive season, which aligns with this study (19). These observations of the android and gynoid regions may provide deeper insight into the core compared with the trunk region alone, as the functional core encompasses the lumbopelvic hip complex, which is not fully included in the trunk region (13,20). These findings suggest that while the upper core (android region) remains stable, the lower core (gynoid region) experiences notable LM losses throughout a competitive season. As the lower core supports stability and the ability to generate force in the upper leg muscles (10,20), these LM losses may translate to worsened performance in CFPs. Future research should investigate the role of LM in DXA-derived android and gynoid regions on the functional capacity of the core.

Limited research exists on total and regional MBR in CFPs, with all current studies utilizing cross-sectional data (5,6). In this study, significant decreases across the season in total and regional MBRs were observed in multiple positions. Total MBR decreased in DBs, DLs, TEs, and WRs, driven largely by losses in LM. Arm LM only significantly changed in the DB group, and arm BMC did not significantly change for any group. However, arm MBR significantly decreased in all position groups except OLs. This illustrates that while LM losses

**Table 7**

**Seasonal changes (mean ± *SD*) in muscle-to-bone ratios by position.*†**

| | DB (*n* = 34) | DL (*n* = 28) | LB (*n* = 25) | OL (*n* = 27) | PK (*n* = 10) | QB (*n* = 10) | RB (*n* = 13) | TE (*n* = 14) | WR (*n* = 27) |
|---|---|---|---|---|---|---|---|---|---|
| Total MBR | | | | | | | | | |
| Pre | 22.08 ± 1.59 | 25.37 ± 1.92 | 23.39 ± 1.58 | 27.34 ± 1.95 | 23.98 ± 2.26 | 22.53 ± 1.30 | 22.87 ± 2.04 | 24.51 ± 1.52 | 21.72 ± 1.13 |
| Post | 21.6 ± 1.68** | 24.94 ± 1.99* | 23.18 ± 1.37 | 27.26 ± 2.06 | 23.51 ± 2.34 | 22.03 ± 1.40 | 22.47 ± 1.85 | 23.74 ± 1.82** | 21.26 ± 1.22* |
| Arm MBR | | | | | | | | | |
| Pre | 18.73 ± 1.20 | 20.08 ± 2.05 | 19.40 ± 1.41 | 20.63 ± 1.98 | 20.33 ± 1.99 | 18.79 ± 1.24 | 19.64 ± 1.55 | 18.92 ± 1.34 | 18.62 ± 1.13 |
| Post | 18.17 ± 1.17*** | 19.48 ± 1.49*** | 18.99 ± 1.42* | 20.41 ± 1.84 | 19.60 ± 1.97** | 18.09 ± 1.15* | 19.06 ± 1.38* | 18.12 ± 1.20*** | 18.14 ± 1.10** |
| Leg MBR | | | | | | | | | |
| Pre | 20.18 ± 1.50 | 22.14 ± 1.96 | 21.14 ± 1.23 | 22.79 ± 1.96 | 20.70 ± 1.60 | 19.95 ± 1.03 | 20.94 ± 1.94 | 21.30 ± 1.70 | 19.96 ± 1.09 |
| Post | 19.67 ± 1.60** | 21.69 ± 2.11* | 20.66 ± 1.12* | 22.43 ± 2.14* | 19.92 ± 1.61** | 19.30 ± 0.81* | 20.32 ± 1.72* | 20.42 ± 1.73*** | 19.34 ± 1.23*** |
| Trunk MBR | | | | | | | | | |
| Pre | 33.60 ± 2.92 | 39.05 ± 3.87 | 35.13 ± 2.97 | 41.70 ± 3.99 | 37.37 ± 4.56 | 34.77 ± 3.43 | 35.41 ± 4.04 | 36.40 ± 2.66 | 32.28 ± 2.73 |
| Post | 32.94 ± 2.77* | 38.63 ± 4.00 | 35.31 ± 2.87 | 41.44 ± 3.34 | 37.41 ± 5.15 | 34.35 ± 4.05 | 35.16 ± 4.34 | 35.87 ± 3.44 | 31.88 ± 2.53 |

*MBR = muscle-to-bone ratio; DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver.

†Significance levels: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.



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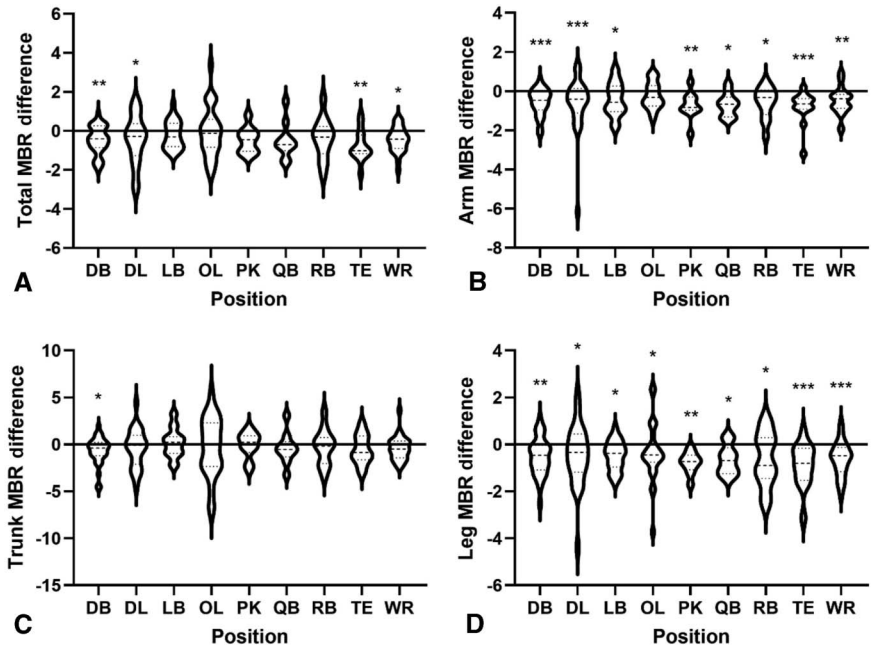


**Figure 3.** Total and regional muscle-to-bone ratio changes from preseason to postseason by position. (A) Total MBR; (B) arm MBR; (C) trunk MBR; (D) leg MBR. DB = defensive back; DL = defensive lineman; LB = linebacker; OL = offensive lineman; PK = punter/kicker; QB = quarterback; RB = running back; TE = tight end; WR = wide receiver. Significance levels: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.

and BMC gains in the arm were insignificant for most groups, the combination of these small independent changes was significant. Similarly, trunk MBR decreased in DBs despite stable LM and BMC. Leg MBR significantly decreased across all position groups, driven largely by LM losses in the leg. These findings suggest that MBR, especially when assessed regionally, may serve as a more sensitive indicator of musculoskeletal balance than LM or BMC alone. Future research should investigate how these MBR changes influence performance and injury risk.

This research has a number of strengths, including a large sample size ($n$ = 188), allowing for position-specific analysis of body composition changes, instead of the broad grouping of positions previously utilized (1,12,16,17,19). In addition, the longitudinal nature of this study shows positional differences in body composition changes, whereas previous research has demonstrated positional differences in regional body composition at single time points (2–4,18,21). The utilization of DXA for measuring body composition allowed for accurate regional subsections, allowing further insights into the longitudinal body composition changes CFPs experience. Furthermore, the retrospective nature of this analysis is strengthened by the stability of the coaching staff, as the head football and strength and conditioning coaches were the same for each season sampled. Finally, the inclusion of player status (injured, did not play, played, and started) as a covariate strengthens the analysis by controlling for competition participation variations.

Despite the many strengths of this study, certain limitations must be acknowledged. First, nutrition, training, and performance measures were not collected, thus limiting the understanding of their influences on body composition. In addition, this analysis covers the changes in body composition that occur in each position across 1 competitive season, limiting understanding of how body composition may change over the course of a collegiate career. Certain demographic variables of interest that may have allowed for this examination were not collected, such as years of experience or eligibility year. In addition, subjects may not have been scanned at the same time of day for their preseason and postseason scans, which may minimally influence composition measures. Finally, although the use of DXA has many strengths, it is costly and has limited accessibility.

## Practical Applications

Findings from this study highlight the need for targeted strategies to mitigate the loss of LM during the competitive season that CFPs face. Although LM declines were commonly observed, the magnitude and region varied by position, emphasizing the importance of individualized, position-specific strength and conditioning programs. This study aimed to provide insight into which positions are most affected in each region. Coaches and performance staff should prioritize lower-body resistance training, particularly for the leg and gynoid/lower core regions. The stability of FM throughout the season may reflect effective existing programming aimed at the maintenance of composition. Continued attention to strategies that prevent FM gains during the season is warranted, with additional emphasis on maintaining LM. The use of DXA to monitor the composition of players during the season may be a valuable tool for coaches to tailor training adjustments to specific positions or individual players. Strength and conditioning professionals, athletic trainers, and nutrition staff can use these findings to inform more precise programming that supports optimal composition of players throughout the season, especially in the later championship phases. Although these insights are specific to DI CFPs, they underscore the broader importance of longitudinal, region-specific body composition monitoring in athlete management.



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